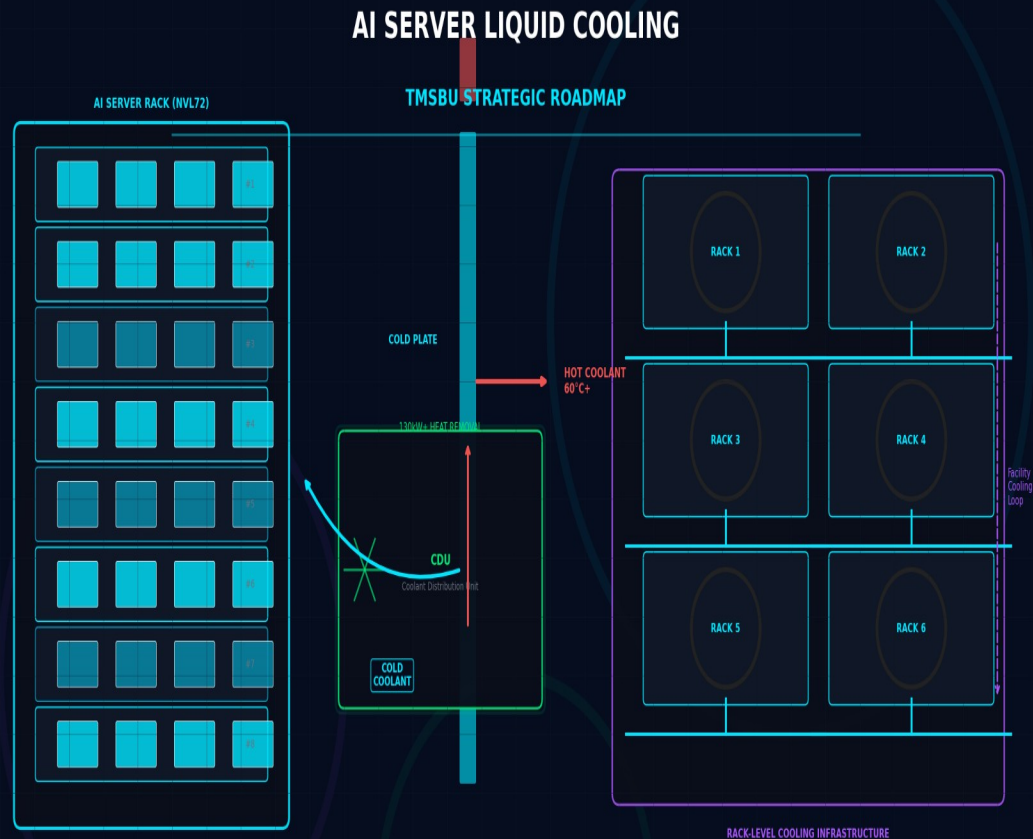


AI Server Liquid Cooling Strategic Roadmap

TMSBU Commercial Opportunity
Analysis



Executive Summary

02

液冷是商業模式轉型，不是產品升級

TREND

1,000W+

Next-gen GPU
thermal load

MARKET

60%*

Liquid cooling adoption
by 2026

*Analyst forecast

HYBRID

AIR + LIQUID

Hybrid architecture
becomes standard

OPPORTUNITY

\$0.8B-\$3.2B

Scenario-based
TMSBU opportunity

ACTION

NOW

6-18M design-in window

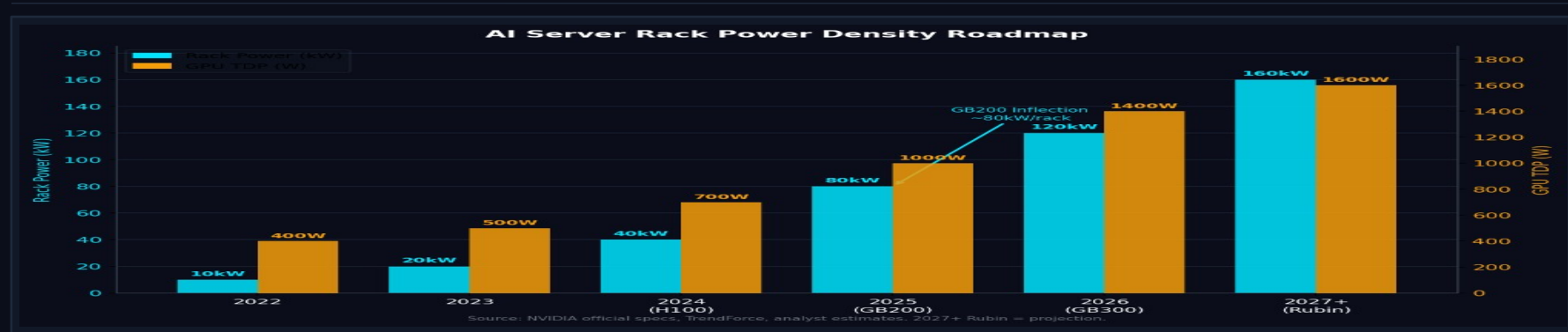
KEY INSIGHTS

- 散熱瓶頸加速浮現：GPU TDP 朝 1,000W+ 發展，傳統風冷實務極限受到挑戰
- 液冷市場 analyst forecast：2026 年 60%+ 新建 AI 資料中心採用液冷
- SOM 為情境假設區間 (\$0.8B-\$3.2B)，需與 TMSBU 實際 pipeline 交叉驗證

Thermal Architecture Shift

03

AI rack 功率密度快速上升，使傳統風冷架構逐步超過實務可承受範圍



1,000W+

Next-gen GPU thermal load
(industry projection)

**100kW+
to 160kW**

Rack power class estimate
(needs validation)

\$39.1B

Liquid cooling TAM
(analyst forecast, 2030)

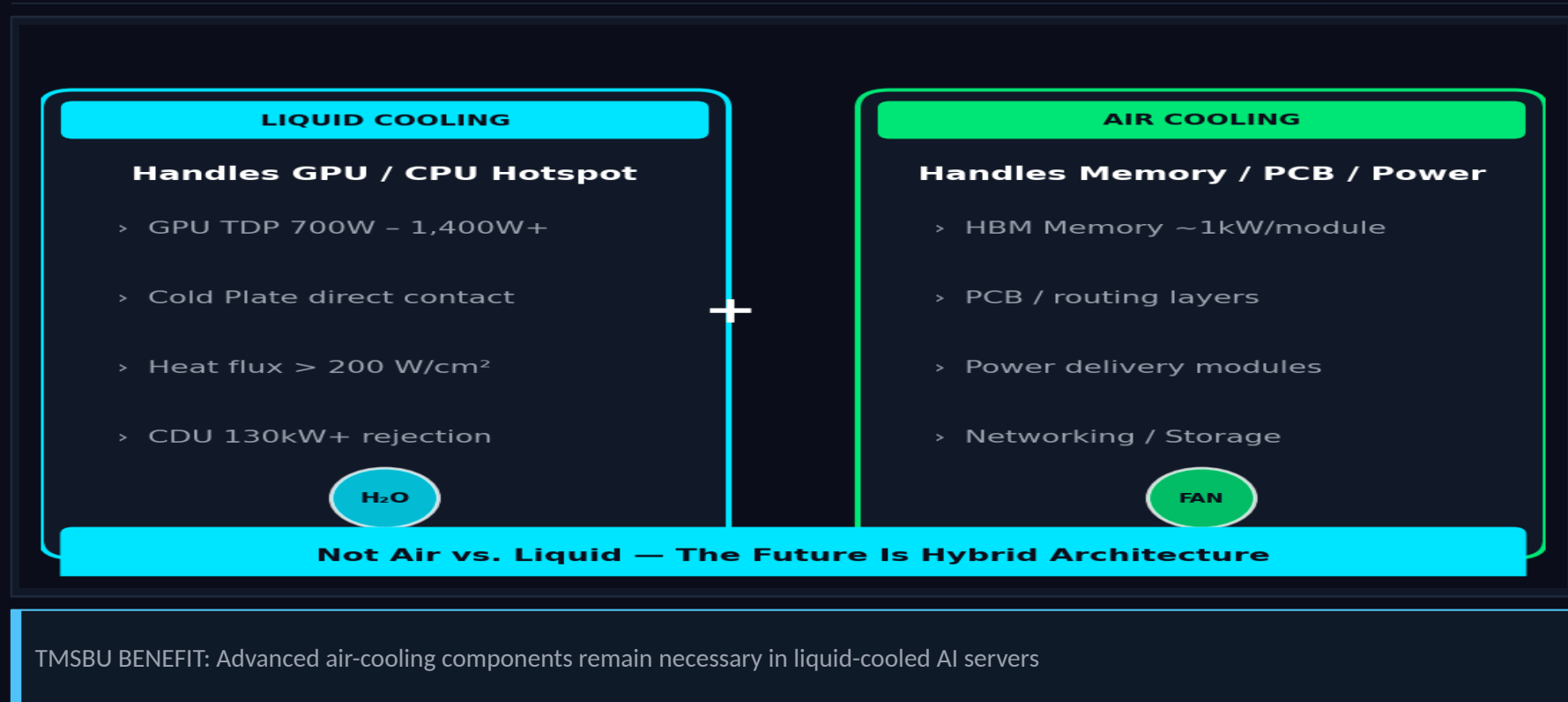
KEY INSIGHTS

- GB200 NVL72 rack 功率進入 80-130kW 等級，推動液冷從選項變為主流方案
- 下一代 GPU (Rubin-era) 功率密度預計挑戰 100kW-160kW 等級，需進一步驗證
- 散熱瓶頸與散熱解決方案供應商的市佔機會高度相關

Why Air Cooling Alone Cannot Keep Pace

04

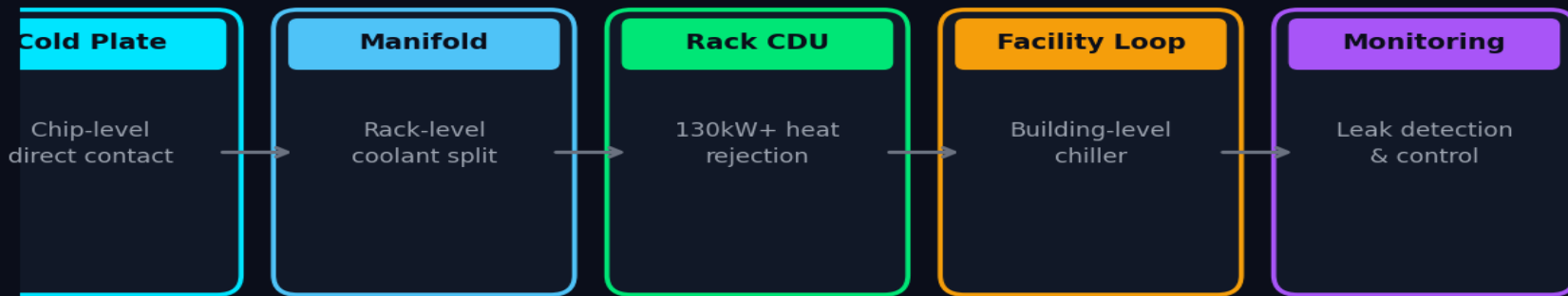
The future is not air vs. liquid — liquid handles hotspots; air remains critical for the rest



Chip-to-Rack Liquid Cooling Architecture

05

System value comes from integration: cold plates, manifolds, CDUs, connectors, monitoring



System integration complexity = highest value source · Vendor complete = highest margin

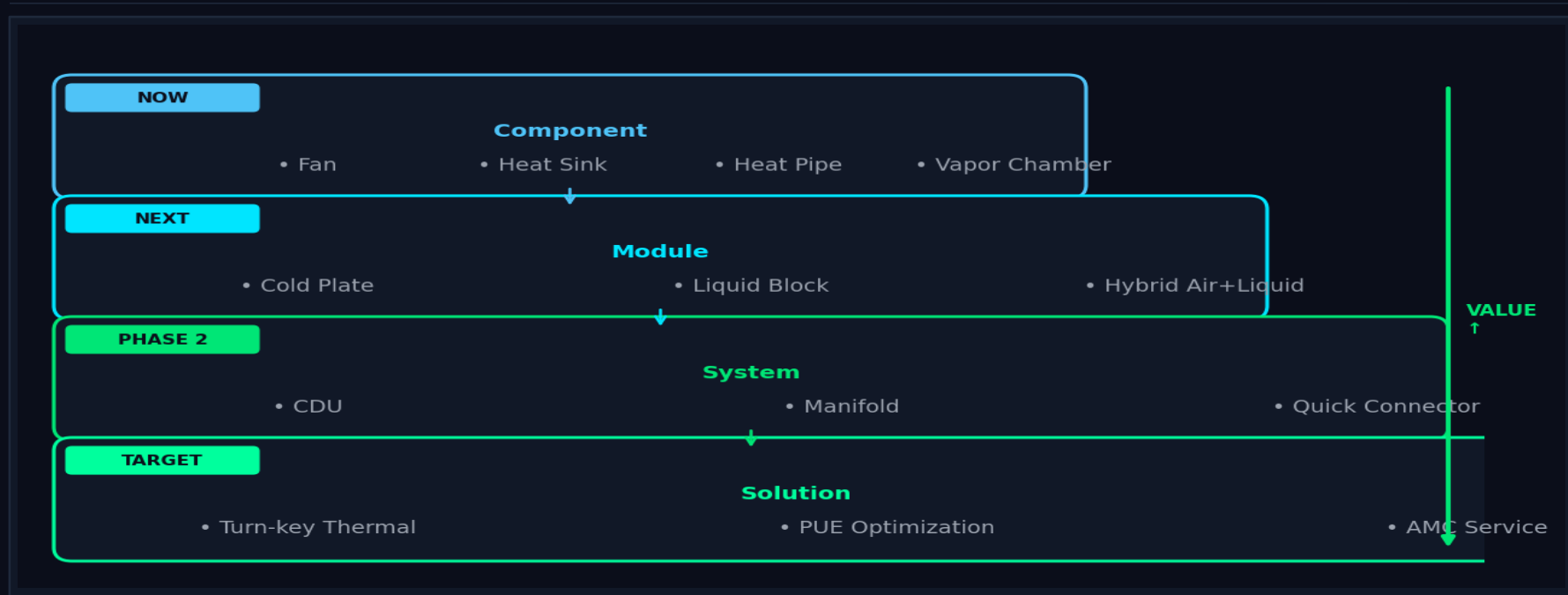
KEY INSIGHTS

- Cold Plate → Server Manifold → Rack CDU → Facility Water Loop → Monitoring / Leak Detection
- 系統整合複雜度是最高附加價值來源 — 不是單一零件供應
- Quick Connector 主要角色是 serviceability (熱插拔、維護)，非熱流主節點

System Solution

06

自然擴張路徑：零件 → 模組 → 系統 → 解決方案



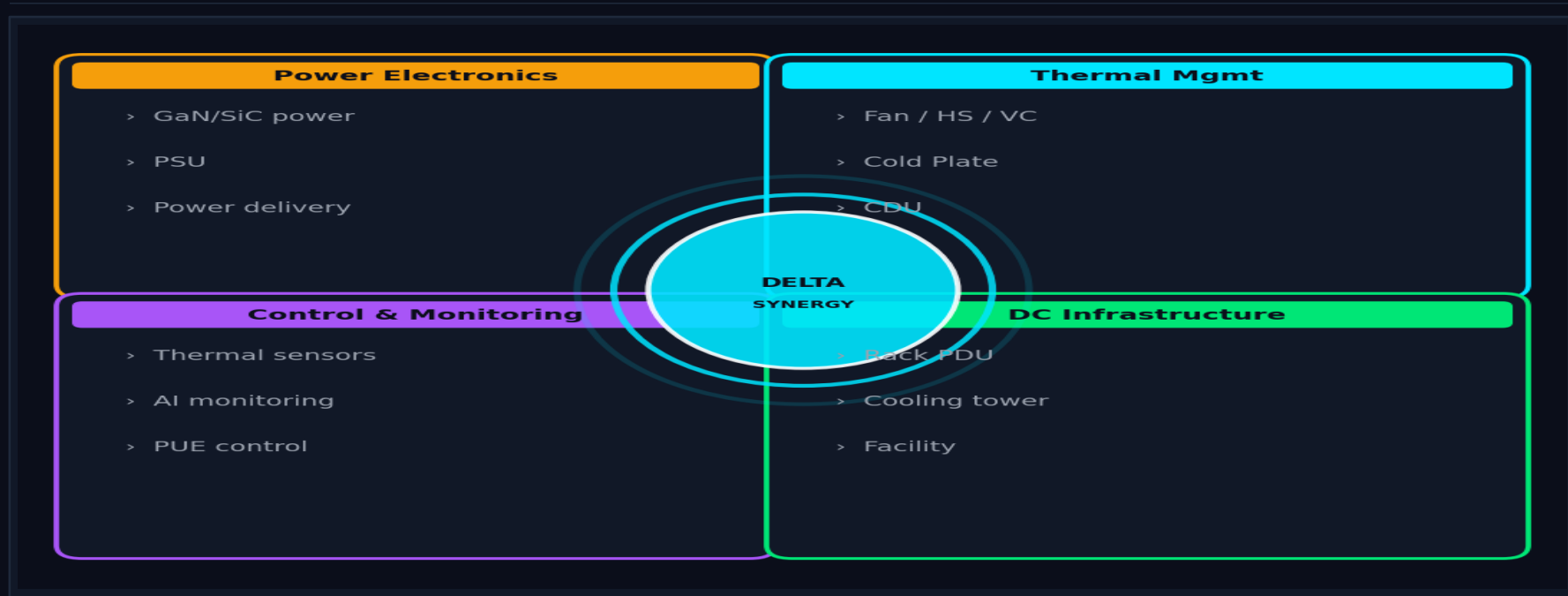
KEY INSIGHTS

- CDU、Manifold、Connector 整合是最高價值延伸區，但可透過自研、合作或併購補齊能力
- 從零件報價升級到系統方案統包報價，改變商業模式與客戶關係
- 目標：成為客戶不可或缺的 Thermal System Partner

Delta Group Synergy: Why TMSBU Can Differentiate

07

Delta 具備將 chip-level 熱設計，延伸整合至 rack-level power、control 與 data center infrastructure 的潛力



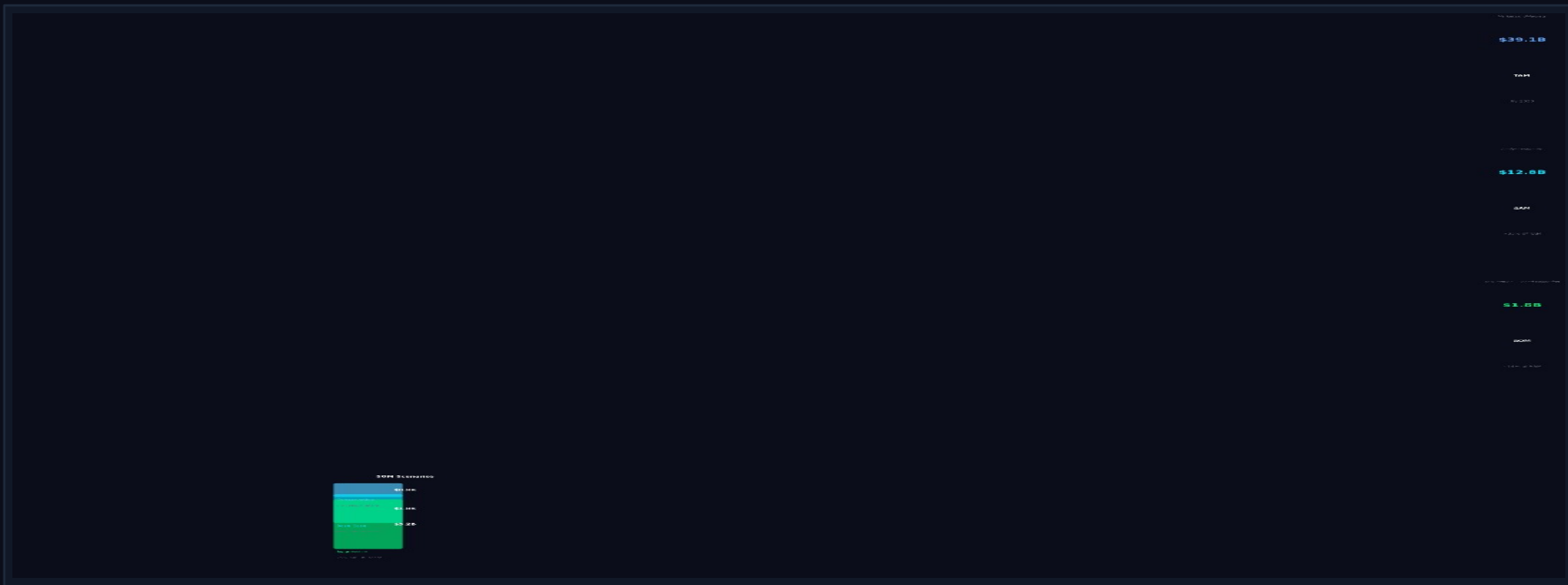
KEY INSIGHTS

- Delta 具備將 chip-level 熱設計，延伸整合至 rack-level power、control 與 data center infrastructure 的潛力
- 相對純散熱供應商，Delta 的集團佈局更難快速被複製
- Manufacturing scale + Global support network 支撐 Tier-1 客戶信任

Market Opportunity: TAM / SAM / SOM

08

市場機會顯著，但 TMSBU 可觸及市場需透過 customer pipeline 與 capability readiness 驗證



KEY INSIGHTS

- TAM \$39.1B (analyst forecast , M&M) : AI 資料中心液冷市場 2030 年規模
- SAM \$12.8B (TMSBU estimate) : 冷板 +CDU+Manifold+Connector 區域
- SOM \$0.8B-\$3.2B (scenario-based assumption) : 需與實際 pipeline 交叉驗證

Capability Gap & Risk Assessment

09

3 項最高優先風險：CDU 策略 / 洩漏可靠性 / 客戶認證週期

Risk Item	Impact	Urgency
CDU Capability	HIGH	HIGH
Leak Reliability	HIGH	HIGH
Customer Qual Cycle	HIGH	HIGH
Quick Connector Qual	HIGH	MED
Coolant Compatibility	MED	MED
Thermal Simulation	MED	MED
Integration Resp.	MED	LOW
Global Support	LOW	LOW

① CDU Strategy

Partner / Acquire / In-house
Decision within 90 days

② Leak Reliability

Quick connector qual
Leak detection mandatory

③ Customer Qual

Start NOW
12-18 month cycles

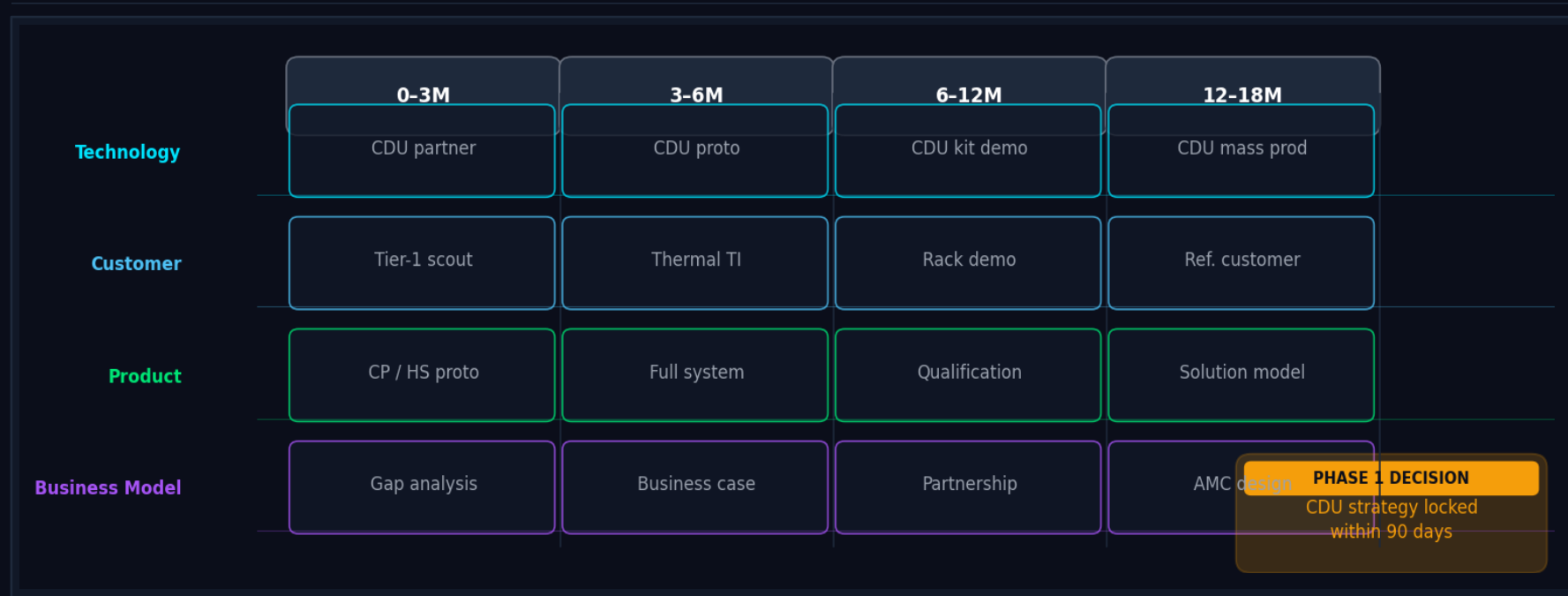
KEY INSIGHTS

- CDU Strategy：Phase 1 最關鍵決策 — 自研 / 合作夥伴 / 併購，90 天內鎖定方向
- 洩漏可靠性是紅線：洩漏事件可造成高價 AI server 損壞與客戶信任風險
- 客戶認證週期 12-18 個月，現在啟動才能趕上下世代平台

Strategic Roadmap & Recommended Actions

10

4 階段 6-18 個月 Roadmap : Phase 1 基礎 → Phase 4 規模化



KEY INSIGHTS

- Phase 1 最關鍵決策：CDU 策略（自研 / 合作夥伴 / 併購），90 天內鎖定方向
- 所有 Phase 終極目標：從 Component Supplier 轉型為 Thermal System Solution Provider
- 延遲代價：錯過 GB300/Rubin 世代早期 design-in 與 reference engagement 機會